

Contents

1 CHAPTER TWO "Text Analysis"	1
1.1 Doubt	1
1.2 Falsifiable	2
1.3 Iteration in gender	11
1.4 Iteration in python	17
1.5 Fluidity	23
1.6 Conclusion	32

1 CHAPTER TWO "Text Analysis"

1.1 Doubt

1

The novel *Orlando: A Biography* (1928), by Virginia Woolf, famously opens with an assertive gender designation followed by an immediate qualification: "He—for there could be no doubt of his sex, though the fashion of the time did something to disguise it" (11).

2

In order to perform quantitative text analysis on this text, the standard tasks of "pre-processing" or "cleaning" would evacuate this sentence of its gender qualification. Cleaning the first sentence not only would strip it of its pronouns, including the gender assertion in the first word, "He." It would also cut the em dash immediately following this "He," signaling the entrance of the narrator and his conspicuous certitude: "—for there could be no doubt of his sex. . . ." Only the following list of words would remain:

3

‘could’, ‘doubt’, ‘sex’, ‘though’, ‘fashion’, ‘time’, ‘something’,
‘disguise’

Text analysis, a process which involves calculating and visualizing textual patterns, requires transforming the source text into a computable format. Cleaning the text strips capitalized words, punctuation, "stop words" (such as articles and prepositions), and inflections in word endings, all of which are deemed to be semantically minor, and wouldn't affect the analysis of more substantial features like nouns, verbs, adverbs, and adjectives. Text analysis requires removing such "minor" elements of language in order to find connections among more apparently substantial terms.

4

The so-called minor elements of the first sentence, particularly the em dash following the "He," betrays the book's investment in gender subversion. As Pamela Caughie points out about the first sentence, "immediately our doubt is aroused. . . The emphasis on an innocent pronoun makes it suspect" ("Double Discourse" 42). Indeed, pronouns cannot be minor in a text about a fictional biography of a 16th-century English nobleman who lives 300 years and undergoes a sex change. *Orlando: A Biography* is a fantastical novel that defies gender and genre expectations. Most famously, the main character himself, Orlando, begins the story as a man and inexplicably changes gender to female about halfway through the novel. Additionally, although hundreds of years pass, Orlando doesn't age, but remains 30 old.

5

This chapter examines how quantitative text analysis works to analyze gender by using Woolf's *Orlando* as a test case. It explores an experimental approach to text analysis that deconstructs gender binaries by drawing connections between computer programming and gender theory. This analysis surfaces a mechanism common to both text analysis and gender performativity, which is *iteration*. There is an iterative method that works on both text analysis and on gender which, I argue, can be harnessed to resignify text analysis procedures. This chapter proposes a text analysis procedure that iterates through various gender terms in *Orlando*, to re-signify their meaning within binary constraints. The goal here is to work within constrictive structures, like the gender binary, in order to unmake them from the inside.

1.2 Falsifiable

1

When I started graduate school in 2016, quantitative text analysis had broken through academic circles into popular discourse. The Literary Studies scholar responsible was Franco Moretti, who introduced many both within and beyond the academy to "distant reading," and published a series of polemical essays in a collection of the same name in 2008. Distant reading, by Moretti's account, involves counting up textual features to derive textual patterns and make inferences. It is an analytical perspective where "distance," he explains, "is a condition of knowledge" (*Distant Reading* 48).

2

A *New York Times* feature of the scholar, released in 2017, helped to propel Moretti and distant reading to audiences beyond the academy. However, shortly after, three former students of Moretti came forward to accuse

him of sexual crimes. The accusations ranged from unsolicited touching to rape. At the time of the alleged incident, one of his accusers was a first-year graduate student.

When I was a graduate student at UC Berkeley in 1984-85, my then-professor Franco Moretti sexually stalked, pressured, and raped me. He specifically said to me, 'You American girls say no when you mean yes.'[...] I reported him to the Title IX officer [...] and he threatened to ruin my career if I pressed charges against him. He said he had powerful friends who were attorneys who would ruin my name. I remained silent for all these years because I was in academia" (Kimberly Latta "Op-Ed: What Happened?" 2017).

When the women came forward, Stanford went no further than claiming they would review the cases. There has been no record of an inquiry, investigation, or any repercussions for Moretti (Liu, Knowles, "Harassment, assault allegations against Moretti span three campuses"). He remains an Emeritus Professor at the same institution.

3

Moretti, at the time, was riding his first wave of academic celebrity. Invited to teach at Berkeley, he had just published a collection of essays, *Signs Taken For Wonders: On the Sociology of Literary Forms* (1983), a book which expressed his already polemical style and sowed the roots of his quantitative approach to studying literature. The first essay, "The Soul and the Harpy/," takes traditional literary criticism as its target, critiquing the methodology of close reading. According to Moretti, close reading is overly subjective, turning the critic to a kind of "Narcissus", "whose only pleasure lay in contemplating his own reflection ("Soul" 14). He proposes an alternative, "falsifiable criticism," a methodology that pursues answers which are irrefutable. He asserts that,

"The day criticism gives up the battle cry 'it is possible to interpret this element in the following way,' to replace it with the much more prosaic 'the following interpretation is impossible for such and such a reason,' it will have taken a huge step forward on the road of methodological solidity" ("Soul" 22).

Moretti here implicitly aligns literary criticism with methodologies from the sciences, methods which are "reproducible."¹ This kind of methodol-

¹Sarah Cohen-Boulakia, Khalid Belhajjame, Olivier Collin, Jérôme Chopard, Christine

ogy squashes the possibility of pesky, alternative interpretations, in favor of conclusions which are “coherent, univocal, and complete” (“Soul” 21).

4

20 years later, falsifiable criticism will find an ideal methodology in quantification. Rather close attention to individual samples, quantification is based on counting patterns across wide swaths of text. Moretti says “we know how to read texts, now let’s learn how *not* to read them” (*Distant Reading* 48). Quantification, as we will see below, simplifies text into workable units which is the basis for higher levels of computation. These simple, workable units enable ambitious interpretations. Moretti explains, “If we want to understand the system in its entirety, we must accept losing something. We always pay a price for theoretical knowledge” (*Distant Reading* 49).

5

As a young graduate student, I found Moretti’s writing both striking and alluring. He is an assertive writer, using mostly short sentences that give his prose a quick rhythm. Despite drawing from disparate fields of knowledge, such as evolutionary theory, systems theory, and Marxism, his arguments move at a light clip, circling methodically around a drain so that, as long as you’re following him, there appears to be no alternative. Paragraphs begin with sentences like,

So, the market selects the canon. (69)

So. As novelistic forms travel through the literary system, their plots are (largely) preserved, while their styles are (partly) lost—and replaced by ‘local’ ones. (134)

What did the birth of a consumer society mean for the European novel? More novels, less attention. (175)

The major metamorphosis of eighteenth-century titles is simple: in the space of two generations, they become much, much shorter. (182)

As the market expands, titles contract; as they do that, they learn to compress meaning; and as they do *that*, they develop special ‘signals’ to place books in the right market niche. (204)

Froidevaux, Alban Gaignard, Konrad Hinsén, Pierre Larmande, Yvan Le Bras, Frédéric Lemoine, Fabien Mareuil, Hervé Ménager, Christophe Pradal, Christophe Blanchet, Scientific workflows for computational reproducibility in the life sciences: Status, challenges and opportunities, *Future Generation Computer Systems*, Volume 75, 2017, Pages 284-298, ISSN 0167-739X <https://doi.org/10.1016/j.future.2017.01.012>.

These sentences, taken from mostly different essays in *Distant Reading*, synthesize arguments and evidence from previous paragraphs. They are more summary of previous points than new argumentation. Yet, their apparent solidity, the lack of a subjective "I" to ground them, strikes me as suspect. It showcases Moretti's characteristic way of transforming inferences, premises, and hypotheses into declarative statements, stating a claim or hypothesis as an objective or self-evident fact.

6

In one of the more famous essays from *Distant Reading*, "Style, Inc.: Reflections on 7000 Titles (British Novels, 1740-1850)" Moretti uses text analysis to study the market for British novels from 1740 - 1850. Here, he explores what titles, which shorten in length over the decades, reveal about the rise and fall of certain genres in the market. He outlines his analytical process:

[F]irst, I describe a major metamorphosis of eighteenth-century titles, and try to explain its causes; next, I suggest how a new type of title that emerged around 1800 may have changed what readers expected of novels; and finally, I make a little attempt at quantitative stylistics, examining some strategies by which titles point to specific genres. Three sections, three pieces in the large puzzle of the literary field. (*Distant Reading* 181-2; emphasis mine)

Moretti's critical method involves posing hypotheses, assembling and analyzing data, making inferences, and occasionally, reframing the original hypotheses. Yet, he repeatedly understates his interpretive moves at each stage of his analysis: "try to explain," "suggest," "make a little attempt." As Stephen Ramsay points out, Moretti presents his own insights as a description of reality, as if "data is presented to us... not as something that is also in need of interpretation" (Ramsay, *Reading Machines* 5). His language not only belies his own role in assembling and interpreting his data, but also his conclusions, for example, that titles function as a proxy for market forces, "a code, in the market: half sign, half ad... where the novel as language meets the novel as commodity" (181). This method misrepresents the role that inferential thinking plays in the admittedly fascinating conclusions he draws, that titles fluctuate in length as a response to market pressures.

7

This faith in the falsifiable—in the power of counting words—infects even those who disavow quantitative methods. A controversial detraction written

by Nan Z. Da claims there is a "fundamental mismatch between the statistical tools that are used and the objects to which they are applied"—that is, to literary objects (620, 601). Yet, despite her oppositional stance toward quantitative tools, Da has something in common with those who champion them: she *agrees* about the objectivity of their procedures. As Ben Schmidt points out, "Far more than anyone I've seen in any humanities article, she asserts that scientists do something arcane, powerful, and true" (Schmidt, "A computational critique of a computational critique of computational critique."). Despite their vastly different views on the role of quantitative methods for studying literature, Da and Moretti appear to agree that these methods ought to provide results that are, at the very least, reproducible.

8

A more precise critique of distant reading comes from Lauren Klein, who deftly argues that it overlooks details which are necessary to cultural critique. In a talk titled "Distant Reading After Moretti," Klein explains,

[I]t's not a coincidence that distant reading does not deal well with gender, or with sexuality, or with race. Gender and sexuality and race are precisely the sorts of concepts that have been exposed and interrogated by attending to non-dominant subject positions. And like literary world systems, or "the great unread," the problems associated with these concepts, like sexism or racism, are also problems of scale, but they require an increased attention to, rather than a passing over, of the subject positions that are too easily (if at times unwittingly) occluded when taking a distant view.

Klein's own work tracing these elements proves how careful attention to what is minor or oblique can surface previously suppressed stories. In one project, Klein uses network analysis methods to bring forth the "ghost" of a previously obscured subject, James Hemmings. Hemmings, a slave who worked for Thomas Jefferson until he could negotiate his freedom, appears in brief references in Jefferson's correspondence. By marking up and visualizing these oblique references, Klein "render[s] visible the archival silences implicit in our understanding of chattel slavery today" (665). Her work demonstrates that the details of Hemmings's life, which are deeply distributed in the correspondence data, requires a kind of reading that is close and careful.

9

By contrast to this close and careful attention, distant reading is permeated by great ambition. In *Macroanalysis: Digital Methods and Literary History* (2013), Matthew Jockers "ask[s] very big questions... regarding

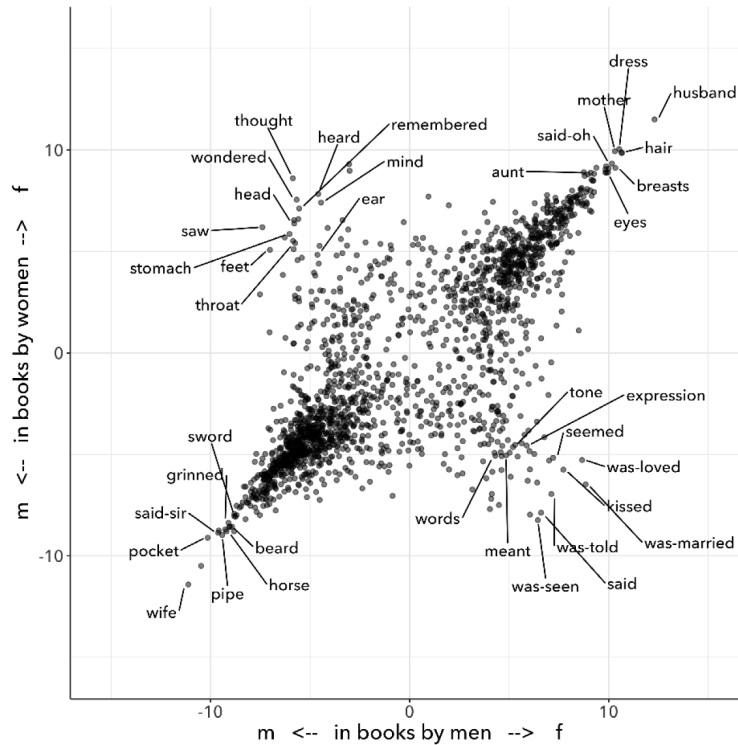
how literary trends are employed over time, across periods, within regions"; In *Distant Horizons: Digital Evidence and Literary Change* (2020) Ted Underwood traces the "larger arcs of literary change that have remained hidden from us by their sheer scale." Most recently, [ADD RECENT EXAMPLES, PREFERABLY MONOGRAPHS?].

10

The ambition to grasp large swaths of literary history may be a proxy for avoiding certain kinds of cultural critique. The proxy functions through an amplification and valorization of what is seemingly objective—for example, the counting of words. With word counts and their resulting graphs and charts doing the work of analysis, one finally closes down the pesky, subjective interpretations associated with close reading. One is finally able to say, once and for all, that "the following interpretation is impossible for such and such a reason" (Moretti, "Soul" 22).

11

There are those, however, who disagree about the apparent objectivity of reproducible methods. Ted Underwood, for example, explains that "algorithms are actually bad at being objective and rather good at absorbing human perspectives implicit in the evidence used to train them" ("Machine Learning and Human Perspective" 92). Underwood creates what he calls "perspectival models" of literary data, where models represent different databases associated with different "perspectives." In one project, on studying gender in novels, Underwood uses a binary classification algorithm to cluster terms that male and female writers, two "perspectives," tend to associate with either gender. From the results, he creates a chart which visualizes the differences between male and female writers and their treatment of gender.



In this chart (see Fig. 1), there are two diagonals. The diagonal moving from the bottom-left to top-right corner represents where the genders agree on gendered terms. The bottom-left shows agreement on terms used to describe the male gender ("pocket", "grinned"), and the top-right shows agreement on the female gender ("eyes," "mother"). The opposite diagonal, running from the top-left to bottom-right corners, shows disagreement. The top-left shows terms that each gender tends to use to describe their own gender ("wondered", "mind") and the bottom-right shows terms that each gender uses to describe the opposite gender ("meant", "kissed"). The chart, according to Underwood, shows two *perspectives* on gender—the male and the female. Rather than making claims to gender in an objective sense, it dramatizes a perspectival shift between the two genders.

12

Underwood's methodology, as he himself admits, reinscribes the gender binary. When explaining the diagram, he predicts that "gender theorists will be frustrated by the binary structure of the diagram. . . I needed a simple picture, frankly, in order to explain how a quantitative model can be said to

represent a perspective" ("Machine Learning and Human Perspective" 98). This reinscription of the "binary structure" happens not only in the choice of model, a classification (i.e. binary) model which makes predictions on a yes/no or, in this case, a male/female scale. It also happens with the choice in visualizing the results in a way that places male and female genders in opposition. The diagonal going from the bottom-left to the top-right, which finds similarity between the genders, reinforces a stereotypical active/passive binary (among other binaries) between men and women. Similarly, the diagonal going from the top-left to the bottom-right, although based on agreement between the genders and their relationships to themselves, nonetheless imposes an *opposition* between how male and female authors describe each other. The graph effectively instantiates its implicit assumption that male and female writers will oppose the other by visualizing this as a feature.

Beyond what Underwood admits, however, is the way the visualization effectively uses the gender binary *as a model* for proving the value of quantitative methods. The gender binary is a necessary presumption in order for the experiment to take place—a ground truth that then makes the choice of a binary-structured model appear appropriate. In order for his diagram to work, gender roles must have been already manufactured into opposition. In other words, binary gender is the model through which his argument about perspectival modeling can make sense.

13

Add Bode's critique.

As Katherine Bode argues, "Underwood's proposal to replace data modeling with sampling and comparison, alone or in concert with perspectival modeling, is ultimately inseparable from his view of the literary past as a single system of interlocking parts. But the literary past, and (quantitative) literary studies, are spaces of highly differentiated knowledges, opportunities, constraints, resources, and possibilities, and these must be considered in any arguments about, or propositions for the future of, the discipline" ("What's the Matter with Computational Literary Studies?" 116).

14

There are valid and even provocative reasons to study gender as a binary system. But the questions that drive such experiments must be differently derived. Questions like "how to distinguish one gender from another gender" effectively reify existing structures of the binary. Alternatively, other work uses text analysis to *deconstruct* social categories. For example, Laura Mandell studies gender through genre. Mandell deploys the popular stylistometry measurement, "Burrow's Delta," which visualizes the "distance" between writing styles by creating branches ("deltas") between different texts.

She finds that Mary Wollstonecraft's style is shared across comparable male writers like William Godwin and Samuel Johnson: "Wollstonecraft's sentimental anti-Jacobin novels most resemble Godwin's sentimental anti-Jacobin novels... whereas her essays most resemble Johnson's writings" ("Gender and Cultural Analytics:: Finding or Making Stereotypes?" par. 29). Her analysis creates new categories that draw gender into genre—squaring the gender binary. She creates doubly refracted categories like, "men writing as men," "women writing as women," "women writing as men," "men writing as women" (par. 35).

15

Applying text analysis to racial categories, Edwin Roland and Richard Jean So explore whether a machine can correctly identify the race of a novel's author. The point, they are careful to say, is not to test whether the machine is correct, but to examine how the categories themselves are constructed. As they explain, "the machine simply measures the robustness of the social constructedness of these given categories and points out what gives them such vigor" ("Race and Distant Reading" 69). Analyzing a large corpus of novels by white and black authors, So and Roland find that black authors generally display more varied vocabulary than white authors. From this result, they infer that White authorship only coheres *as a category* against the variance of Black authorship. In other words, White authorship is only visible as a category when it is compared to Black authorship, which displays overall more varied uses of language forms.

Interestingly, this quantitative exercise points Roland and So toward a peculiarity in the results: that the algorithm wrongly categorizes Black-authored novel, James Baldwin's *Giovanni's Room* (1956), as being written by a White author. This misclassification is due to a single word—"appalled"—whose usage, according to the model, has generally been a signal of White authorship. The term only occurs once in the text, in an early scene where the narrator David describes his strained relationship to his father: "I did not want to be his buddy. I wanted to be his son. What passed between us as masculine candor exhausted and appalled me" (Rpt. in Roland, So 71; emphasis mine). Given the connotation of whiteness in "appalled" and its middle French root, "apalir" (to grow pale), Roland and So explain that this term—which emerged as a result of the computer's mistake—enables their analysis to coordinate gender to race: "the moment David develops a troubled relationship to normative masculinity [as] also the moment he becomes 'White'" (71). Rather than reify racial categories, this computational experiment points out that they are constructed on certain assumptions.

16

So and Roland began with the binary form, the binary categories of Black/White, but they ended with its fracturing. This fracturing, evident in the computer's misclassification of authorship in *Giovanni's Room*, leads to a reading which coordinates race with gender and sexuality. This "queering the machine's color line," the authors explain, opens up new readings of intersectionality in the novel which "in their current form, our data and model are not robust enough to handle" (72). Rather than reproducible, this method is fracturing and refractive, both reflecting and corrupting the constructed nature of binary structures back at the critic. It brings us closer to the methodology that I outline in the next section, *iteration*.

1.3 Iteration in gender

1

Multiple text analysis projects draw from Butler's concept of performativity. In particular, they take the language of performance and qualities of the performative to describe and conceptualize their methodologies. For example, Mandell's project, introduced above, takes the behavioral quality of gender performativity and associates it with authorial style, asserting that both gender and genre "are... highly imitable," so that "anyone can adopt gendered modes of behavior, just as anyone can write in genres stereotypically labeled M/F" (par. 30). Bode applies the "performative" to illustrate the analytical act itself, which necessarily intervenes and becomes inseparable from data as object of analysis, where "performativity" is "attuned to the coconstitution of computational methods and objects" ("What's the Matter with Computational Literary Studies"). Her project, *Reading at the Interface*, explores how "meaning is not carried by texts but produced in situated interactions between readers, texts and contexts," including computational tools.² Another project, from the field of Computational Humanities, Kent K. Chang, Anna Ho and David Bamman cite Butler's performativity as motivation to computationally "measure subversion" in TV show dialogue. Their uses a language model to predict whether samples of a TV show, *The Big Bang Theory*, are queer-coded. If the model marks certain sections of dialogue between heterosexual characters as queer, they argue, then it will have indicated moments of "subversion" in the show (Chang et al.).

2

Unlike those projects, however, this project draws from performativity's

²Bode, Katherine. "Data beyond representation: From computational modelling to performative materiality." MLA Convention, Seattle, 10 January 2020. <https://katherinebode.wordpress.com/home/mla-convention-2020/>

methodology in depth. In doing so, it demystifies a common misreading of performativity—that it denotes an act or series of acts that can be imitated at will. Indeed Butler's follow up to *Gender Trouble: Feminism and the Subversion of Identity* (1990), aimed to clarify how performativity is bound by social systems and norms in ways that delimit agency. In this work, *Bodies that Matter: on the Discursive Limits of Sex* (1996), Butler explains that gender performativity is a compulsory process that constitutes the subject. In Butler's reasoning, the body only materializes (*matters*) through the repetition of gender norms. And it is through repetition that the body also assumes subjectivity. Crucially, "there is no subject who decides on its gender. . . . on the contrary, gender is part of what decides the subject" (ix). In other words, gender is an activity that determines both the body and subjectivity. It is from this context of a compulsory gender system where the subject must find a way to re-work the terms of expression.

3

Butler's argument for performativity draws from poststructuralist feminist theorist Luce Irigaray. Irigaray is best known for her readings of phallogocentrism (the idea that the phallus is a focus or lense for understanding other phenomena), in the works of classical and modern philosophers, like Plato, Aristotle, Descartes, and Kant. Irigaray finds that these philosophers represent the feminine through masculine terms, "on the basis of masculine parameters," as she describes in her book, *The Sex Which Is Not One* (Irigaray 23). Her conceptualization of the feminine's circumscription within the gender binary directly influences Butler's conceptualization of performativity and how it might resist rigid signficatory systems (18).

4

Irigaray's analysis of the feminine centers on its unrepresentable status within dominant framings like the gender binary. She argues that the binary of male/female effectively *precludes* the representation of the feminine: "The economy that claims to include the feminine as the subordinate term in a binary opposition of masculine/feminine excludes the feminine, produces the feminine as that which must be excluded for that economy to operate" (Butler 10). While the feminine is effectively excluded from the system, it conditions the representation of terms *within* the system. This excluded feminine, this "constitutive outside," makes possible the creation of the inner feminine, the "domesticated" feminine. It makes legible an internal feminine. She, what Butler calls the "unspeakable" feminine, is the *matter* from which the binary may come into intelligibility.

5

This external feminine is *matter*: the generative, creative material from

which distinctions emerge. Butler explains that,

every explicit distinction takes place in an inscriptional space that the distinction itself cannot accomodate. Matter as a *site* of inscription cannot be explicitly thematized. And this inscriptional site or space is, for Irigaray, a *materiality* that is not the same category of "matter" whose articulation it conditions and enables... this second inarticulate "matter" designates the constitutive outside of the Platonic economy; it is what must be excluded for that economy to posture as internally coherent" (emphasis original, 12-13).

Irigaray's association of this excessive feminine with "matter" draws simultaneously from a classical analogy between the form/matter and masculine/feminine, in the Aristotelian tradition, and from the etymological association between the feminine as *mater*, mother, and significantly, as *matrix*, the generative material for all life. It is this *matrix*, this "inscriptional space", which cannot be articulated, but from which all distinctions of male/female can emerge.

6

How is this feminine to enter into representability? How can it be invoked directly without subscribing itself to the ruling structure? Or, in Butler's formulation, "how can one read a text for what does not appear within its own terms?" (*Bodies* 11). As a constitutive exclusion which makes the intelligible binary possible, the excluded feminine cannot be represented in the same terms (10).

7

The answer, as it turns out, involves breaking open signifiatory procedures. Butler considers the referential power of language: "Can language simply refer to materiality, or is language also the very condition under which materiality may be said to appear?" (6). Butler offers the example of the physical body. In order to refer to a physical body, Butler explains, language must first *assume* a body. That is, the body must first be posed in language. Therefore, Butler reasons, "signification produces as an effect of its own procedure the very body that it nevertheless and simultaneously claims to discover as that which precedes its own action" (6). In their characteristically clausey prose, Butler explains that signification creates the body which it appears to reference. Which leads them to a major realization: "the mimetic or representational status of language... is not mimetic at all. On the contrary, it is productive, constitutive, one might even argue performative" (*Bodies* 6).

Performativity, then, denotes the perpetual creation and recreation of norms, a discursively-based, signficatory cycle which *produces* that which it appears to reference. As they explain, "performativity must be understood not as a singular or deliberate 'act,' but, rather, as the reiterative and citational practice by which discourse produces the effects that it names" (xii). According to Butler, it is through this discursive performativity that "sexed" bodies emerge: "'Sex' is not a simple fact or static condition of a body, but a process whereby regulatory norms materialize 'sex' and achieve this materialization through a forcible reiteration of those norms" (xii). What appears as physical or biological category, "sex," is effectively delimited by and through a discursive framing. Bodily materiality is a "sedimentation" formed through repetition, whereby meaning is signified and re-signified endlessly, and never completely.

Categories like sex and gender, as Butler explains, are never stable, but require constant repetition:

This "being a man" and this "being a woman" are internally unstable affairs. They are always beset by ambivalence precisely because there is a cost in every identification, the loss of some other set of identifications, the forcible approximation of a norm one never chooses, a norm that chooses us. (86)

Repetition stabilizes gender identity against a field of potential exclusions, what Butler calls the "cost in every identification." Every identification creates this field of exclusion. And it is this material which will come back to destabilize coherence. With an ominous overtone, Butler explains that, "The political terms that are meant to establish a sure or coherent identity are troubled by this failure of discursive performativity to finally and full establish the identity to which it refers" (Butler 140).

This necessity to repeat gender signification is a weakness, which Butler will use as an exploit. For, within this signficatory circle, lies the possibility of *resignifying* terms. Resignification happens by repeating what has been excluded from the signficatory economy, "citing" what Butler calls a "repudiated" meaning. As Butler explains, every "identification takes place through a repudiation which produces a domain of abjection (xiii). This domain of abjection, containing identities that are repudiated, offers "a field of disruptive possibilities" for resignification (Butler 10). Butler explains, "this

disavowed abjection will threaten to expose the self-grounding presumptions of the sexed subject" (xiii). Through this threat, "what has been foreclosed or banished from the proper domain of "sex"... might at once be produced as a troubling return" (Butler xxx).

11

If the troubling return is repeated continually, Butler argues, the stability of the original identity will be brought into question. Butler offers an example with the term "queer," which is a term that harnesses its own history of repudiation for political purposes, "resignifying the abjection of homosexuality into defiance and legitimacy (Butler xxviii).

The compulsion to repeat an injury is not necessarily the compulsion to repeat the injury in the same way or to stay fully within the traumatic orbit of that injury. The force of repetition in language may be the paradoxical condition by which a certain agency—not linked to a fiction of the ego as master of circumstance—is derived from the impossibility of choice. (183)

It is through the force of repetition that repudiated meanings, which necessarily draw from insulting and painful pasts, come to alchemize new significations. The paradox is that what appears to be a cyclical or even robotic behavior takes on an agentic capacity. Automata, at some point, attain agency.

12

Subversion thus reaches toward a destabilization of the ruling system. And this is where Butler's theory is often misunderstood. For example, Chang et. al's text analysis project claims that: "gender performativity provide[s] the essential framework for understanding how characters on TV subvert traditional gender roles and expectations" (4). The authors argue that in the TV show *Big Bang Theory*, "subversion" in the TV emerges in "an undercurrent of emotional intimacy and dependency that challenges the rigid boundaries of what is considered acceptable" between two of the straight male characters, Raj and Howard, who display a comically close and intimate relationship (Chang et al., 10). In their conclusion, they posit that, "Raj and Howard offer more than occasional punchlines; they represent an instance of queer love: they have to invent a form of togetherness for themselves" (Chang et al., 2024, p. 9).

However, according to Butler, subversion is more than exhibiting a non-dominant norm. It is doing so in a way that challenges established norms. Butler carefully stipulates against instances of apparent subversion which effectively prop up heterosexual regimes:

[T]here are forms of drag that heterosexual culture produces for itself—we might think of Julie Andrews in *Victor, Victoria* or Dustin Hoffmann in *Tootsie* or Jack Lemmon in *Some Like It Hot* where the anxiety over a possible homosexual consequence is both produced and deflected within the narrative trajectory of the films. These are films which produce and contain the homosexual excess of any given drag performance, the fear that an apparently heterosexual contact might be made before the discovery of a nonapparent homosexuality. This is drag as high het entertainment, and though these films are surely important to read as cultural texts in which homophobia and homosexual panic are negotiated, I would be reticent to call them subversive.
(85)

Despite being queer-coded, these characters remain within heterosexual relationships. As a result, their queerness effectively "produce[s] and contain[s] the homosexual excess" within the parameters of the show. They use humor to temper these excesses. Borrowing from Irigaray, they are an example of "domesticated" queerness, rather than subversive queerness.

13

Subversion aims toward destabilization. In doing so, repetition is key, enabling the introduction of the repudiation back into the system. Returning to Irigaray's writing, Butler explains that Irigaray achieves subversion by "mim[ing] philosophy... and, in the mime, tak[ing] on a language that effectively cannot belong to her" (12). Butler cites a passage where Irigaray *mimes*, or repeats, the language of Plato. Her repetition, however, has a particular effect: she "performs a repetition and displacement of the phallic economy" (Butler 18). In an interesting double-level inhabitation, Butler approximates what Irigaray must be feeling to inhabit Plato's written style:

I will not be a poor copy in your system, but I will resemble you nevertheless by miming the textual passages through which you construct your system and showing that what cannot enter it is already inside it (as its necessary outside), and I will mime and repeat the gestures of your operation until this emergence of the outside within the system calls into question its systematic closure and its pretension to be self-grounding." (*Bodies* 18)

Irigaray undermines the authority of the masculine through repetition, by "cit[ing] Plato again and again, but the citations expose precisely what is

excluded from them, and seek to show and to reintroduce the excluded into the system itself" (Butler 18). Butler explains that "This is a taking of his [Plato's] place, not to assume it, but to show that it is *occupiable*... a kind of overreading which mimes and exposes the speculative excesses in Plato (emphasis original, 11).

14

Through repetition, Irigaray displaces the logic of phallogocentrism, introducing something external to the system while simultaneously remaining within its terminology. Deception emerges from resemblance, and insubordination through subservience. The key is repetition, or more precisely, what Butler refers to as *iteration*, a continual miming of the authorizing norm. Through the iterative process, one brings back again and again the repudiated meaning as a troubling return. "The task," according to Butler, "will be to consider this threat and disruption... as a critical resource in the struggle to rearticulate the very terms of symbolic legitimacy and intelligibility" (xiii).

1.4 Iteration in python

1

Iteration is a core concept in programming. Iteration means repeating the same action to each bit of data in a collection of data.

4

The looping mechanism quickly cleans a text and remove extraneous or undesirable elements analysis. Elements that are highly frequent, like articles, prepositions, and pronouns, can drown out other terms which are considered more substantial, like nouns and verbs. Additionally, irregular cases also need to be standardized, because capital letters would count toward different words (for example, "talk" and "Talk" would be considered two different words). For similar reasons, word endings and inflections like plural forms (endings in "s") and tenses ('walked', 'walks' or 'walking') are converted into a root form. Once the text has been regularized, then the counts will be more accurate to its content.

Using a loop, a person may *iterate* through each word in a text, check whether it needs to be "cleaned," and then perform a set of cleaning actions. For example, a first round of cleaning might check for the presence of punctuation and capital letters. Below is an example of a **for loop** that filters out punctuation and transforming any capital letters into lowercase forms:

```

more time to run.
    clean = []
    for word in text:
        if word.isalpha():
            clean.append(word.lower())

```

3

The loop begins by creating an empty placeholder, `clean`, where words will be saved in their clean form. after passing the filters below. The next line begins the for loop, which iterates through each word in the text, in our case, in Woolf's novel, *Orlando*. The third line contains the first filter, `isalpha()`, which checks if the word is comprised of only alphabetic characters (i.e. it is not a number or punctuation). If the word is alphabetic, it passes the filter. The word moves to the next line, where it is lowercased, `lower()` and finally added to `clean`. When the loop is done running over every word in the text of *Orlando*, the final list of words in `clean` will only contain alphabetic and lowercased letters.

5

The next task in cleaning is to remove what are called **stops** or "stop words," which consist of prepositions, articles, pronouns, and auxiliary verbs.

```

["i", "me", "my", "myself", "we", "our", "ours", "ourselves", "you",
 "your", "yours", "yourself", "yourselves", "he", "him", "his",
 "himself", "she", "her", "hers", "herself", "it", "its", "itself",
 "they", "them", "their", "theirs", "themselves", "what", "which",
 "who", "whom", "this", "that", "these", "those", "am", "is", "are",
 "was", "were", "be", "been", "being", "have", "has", "had", "having",
 "do", "does", "did", "doing", "a", "an", "the", "and", "but", "if",
 "or", "because", "as", "until", "while", "of", "at", "by", "for",
 "with", "about", "against", "between", "into", "through", "during",
 "before", "after", "above", "below", "to", "from", "up", "down", "in",
 "out", "on", "off", "over", "under", "again", "further", "then",
 "once", "here", "there", "when", "where", "why", "how", "all", "any",
 "both", "each", "few", "more", "most", "other", "some", "such", "no",
 "nor", "not", "only", "own", "same", "so", "than", "too", "very", "s",
 "t", "can", "will", "just", "don", "should", "now"]

```

Due to their high frequency and low semantic value in comparison to nouns, verbs, and adjectives, and adverbs, stop words will skew the results of analysis, so they must be removed.

```
no-stops = []
for word in cleaned:
    if word not in stops:
        no-stops.append(word)
```

6

The loop takes each word the `clean` list, and checks to see if that word is also contained in above list of `stops`. If the word is not a stop word, then it will be added to a new list, `no-stops`. After this processing, the following words remain from the first sentence of Orlando:

```
['could', 'doubt', 'sex', 'though', 'fashion', 'time', 'something',
 'disguise']
```

7

As mentioned in this chapter's introduction, the gender pronouns, "He" and "his", as well as the em dash following "he" has been removed.

The final step of cleaning involves stripping word inflections to get the word's root form, or "lemma". This process involves looking up each word, one by one, to return the shortest lemma found. It uses the WordNet database, an electronic lexical database used for computational linguistics. WordNet contains every word's root form, as well as "synsets" or synonym sets, representing underlying relationships between words.³ In the python code, the lemmatizer function `lemmatize()` is called from the `WordNetLemmatizer` module.

```
lemmatized = []
for word in no-stops:
    lemma = WordNetLemmatizer.lemmatize(word)
    lemmatized.append(lemma)
```

8

The above code iterates through the list of `no-stops` (which has already been stripped of punctuation, numbers, capitalization, and stop words) and lemmatizes each individual word using the `lemmatize()` function. It saves the word to a new variable, called `lemma`, which it finally appends to the list called `lemmatized`. After iterating through every word in the `no-stops` list, and lemmatizing each one by one, the text is ready for analysis.

³Christiane Fellbaum (1998, ed.) WordNet: An Electronic Lexical Database. Cambridge, MA: MIT Press.

Text analysis is, at a most basic level, about counting words. From counting how frequently a word appears, one can start to derive patterns about word usage. The text analysis library Natural Language Toolkit, or NLTK, is a python library, or collection of code, written specifically for working with language data. Due to its modular structure, it's a good library for beginners in Python (it was the first library that I ever learned). NLTK comes packaged with useful functions for simple text analysis tasks, like `q concordance()`, which displays the context window surrounding a given word, and `similar()`, which finds words that are used in similar contexts to a given word.

For example, running `concordance()` on the word "woman" will output sentences that contain that word from *Orlando*:

Displaying 15 of 187 matches:

```
ty manly charm quality old woman loved failed growing old w
en without knowing perhaps woman heart intricate ignorance
indow pulled among cushion woman laid worn old made bury fa
red dyed cheek scarlet old woman loved queen knew man saw o
finger swore vilely master woman scarcely le bold speech le
ment panic twelve poor old woman parish today drink tea ton
perverse cruel disposition woman broke engagement night eve
verladen apple old bumboat woman carrying fruit market surr
embassy figure whether boy woman loose tunic trouser russia
ed stared boy ala boy must woman could skate speed vigour s
me standstill handsbreadth woman orlando stared trembled tu
d asked tumult emotion old woman answered skin bone trulls
and longer full prying old woman said stared one face bump
save sea bird old country woman hacking ice vain attempt d
ce melt heat pity poor old woman natural mean thawing must
```

10

This context window reveals a text that has been stripped of punctuation, capital letters, prepositions, articles, pronouns, auxiliary words. What remains is largely nouns, verbs, and adverbs. The task is to then count and analyze these remains.

11

For that purpose, another function, called `similar()` displays words that are used in similar contexts of the target word. This function takes a canonical concept from NLP called "distributional similarity," where words that appear in similar contexts tend to have similar meanings. For woman, those words are the following,

```
['man', 'moment', 'night', 'boy', 'word', 'world', 'child', 'pen',  
'ship', 'door', 'one', 'room', 'window', 'light', 'little', 'lady',  
'table', 'book', 'queen', 'king']
```

To compute the results of `similar()`, the program first takes the context of the target word from `concordance()`. Then, it counts each of the words surrounding woman. As seen above with the `Counter()`, it keeps a running tally of these words. Then, it searches other words in the database and, again, takes a count of surrounding words for these words. It then looks for words that have the most shared contexts, that is, the most shared words within their context window, with the word woman. Finally, it outputs a list of terms that share the most contexts with the word.

12

For the program then, word meaning is computed through word counts. Importantly, the computer does not understand or impute *meaning* to the words. Rather, it only counts each word as a string, that is, as a piece of data composed of alphanumeric sequences. From there, it derives meaning from patterns of usage.

13

That is, in a nutshell, what so-called *deep learning* methods do. Deep learning methods, like those that drive generative text processes (discussed in detail in chapter 5) move from displaying the results of word counts into generating word predictions.

Many of these methods are based on the concept of "word embeddings" or "word vectors" to ascribe machine-interpretable meaning to strings. Like `similar()` and `concordance()`, word embeddings work from patterns of word similarity based on context. Unlike the previous methods, however, word embeddings encode precise numeric values to a given word based on its context.

14

A vector for a single word, "woman," like the one below, will contain a list of numbers that represent a high similarity score between "woman" and another word.

```
['child', 0.9371739625930786,  
'mother', 0.9214696884155273,  
'whose', 0.9174973368644714,  
'called', 0.9146499633789062,  
'person', 0.9135538339614868,  
'wife', 0.9088311195373535,
```

```
'being', 0.9037441611289978,  
'father', 0.9028053283691406,  
'guy', 0.9026350975036621,  
'known', 0.8997253179550171,  
...]
```

Based off a popular model trained from twitter posts, the vector which represents “woman” contains a list of numbers that score the similarity of the word “woman” to other words in the dataset. Here, the word “woman” is most closely associated to the word “child,” with a similarity score, or “weight,” of .93, or 93%, then with “mother,” with .92, then “father,” with .90, and so on.

15

Commonly, word vectors are organized into a matrix, or tabular, format. Using this matrix format, further mathematical operations are possible using statistics, linear algebra, and calculus—the building blocks of deep learning methods.

The inventors of word vectors introduced their power to the world via a formula that uses math to compute language meaning (Mikolov et al. 2013). The famous formula uses a mathematical equation to calculate the meaning of the word "Queen". Here, the vector for the word "queen" is calculated by subtracting the vectors of "man" from "king," and finally, adding the vector for "woman."

$$\text{vector("Queen")} = \text{vector("King")} - \text{vector("Man")} + \text{vector("Woman")}$$

The cross-cultural intelligibility of this formula, which introduced word vectors to the world, cannot be understated. Gender, which is necessarily imbricated with imperialist concepts, becomes an operand for computing language meaning. Like Ted Underwood’s graph which puts gender into an oppositional relationship, gender is the *model* which shows the algorithm to be true.

They are called "vectors" because these numeric values represent points in a graph. By graphing word similarity, word vectors translate relationships of word context into geometric space.

Like Butler’s matter or *matrix* which produces the binary of masculine/feminine, here is another matrix that produces meaning according to masculinist parameters. The matrix format of the word vector enables a host of mathematical operations and algorithms: from example, cosine similarity calculates the distance between two vectors, that is, two words; and

regression analysis guesses the relationship between an independent and dependent variable, or relation between one word and another.

16

In the following section, I use these word vectors as a starting point to explore terms related to gender terms in *Orlando*, starting with the terms “woman” and “man.” Although I start with the gender binary, word vectors enable me to quickly proliferate beyond it. Rather than resolving gender into an oppositional structure or a neat formula, it cracks open.

1.5 Fluidity

1

But it is more than Orlando the character who defies realist expectations. As Woolf scholar Pamela Caughie puts it, “The text of *Orlando* is as unstable as the sex of Orlando” (42). Subtitled *A Biography*, *Orlando* draws from English takes real life as its source material—except it re-works this material into what critics Christy Burns and Jane de Gay describe as parody and satire.⁴ First, English literary history is reproduced and satirized, with each chapter of the novel taking different periods like the Elizabethan, Restoration, and Victorian periods, and featuring (and mocking) characters like Queen Elizabeth and Alexander Pope. Moreover, the model for Orlando’s character comes from Virginia’s friend and lover, Vita Sackville-West.⁵ Episodes from Sackville-West’s real life, like her heartbreak with the author Violet Trefusis and her disinheritance of the family estate at Knole are represented in Orlando’s heartbreak with Sasha and (after her transition) her legal battles over her property and titles. According to critics

The most unstable element of the text, however, is the language, whose experimental quality evokes its peers in the early modernist canon. According to Caughie,

In its rhetorical transports, Woolf’s novel challenges the reference theory of meaning. In particular, it questions the notion **that words get their meanings from things they refer to**; the definition of words and categories by their essential traits; and

⁴Burns, Christy L. “Re-Dressing Feminist Identities: Tensions between Essential and Constructed Selves in Virginia Woolf’s *Orlando*.” *Twentieth Century Literature*, vol. 40, no. 3, 1994, pp. 342–364. **JSTOR**, www.jstor.org/stable/441560; De Gay, Jane. “Virginia Woolf’s feminist historiography in *Orlando*.” *Critical Survey*, vol. 19, no. 1, 2007, p. 62+.

⁵“‘If I Saw You Would You Kiss Me?’: Sapphism and the Subversiveness of Virginia Woolf’s *Orlando*.” *PMLA*, vol. 103, no. 1, 1988, pp. 24–34. www.jstor.org/stable/462459

the isolation of words and statements from their contexts of use in order to interpret them. (Caughie, "Double Discourse, 44-45).⁶

I agree with Caughie that language's referential power is a major, if not the central, problematic in *Orlando*. However, unlike Caughie, who positions the main character's gender fluidity as the cause of language's instability, I explore the opposite movement: following Butler, this reading poses language forms as determining and producing gender norms and their re-signification.

2

Crucially, for Butler, re-signification works through iteration, which is repetition with a difference. For the following close reading, I pursue an *iterative* text analysis method that runs the same quantitative computation over and over again, with slight difference: it intersperses quantitative analysis with close reading, dipping in and out of the text to examine words in their original context. This toggling between close and distant views of language in the novel recalls a procedure that Andrew Piper calls "bifocal" reading, which "explicitly constructs the context through which something is seen as significant" (17).⁷ Unlike Piper, however, who uses the quantitative component to "cede" "some measure of subjectivity" to the critical process, this method explicitly embraces the critic's inclinations as the driving force of analysis. Inclination, including those motivated by pleasure and surprise, will be essential to divert the expected path of gender's supposed reproducibility.

3

I begin with two gender terms from the novel, "woman" and "man." Using word vectors, and the `Word2Vec`,⁸ library in python, I compute words used in similar contexts. The function to compute similar words, written below, uses cosine similarity to calculate the graphical distance between two words, represented as vectors. In other words, the function searches for words that appear with similar coordinates to "woman" and "man" in vector space.⁹

⁶Pamela Caughie. "Virginia Woolf's Double Discourse." *Discontented Discourses : Feminism/textual Intervention/psychoanalysis*. Ed. Marleen S. Barr & Richard Feldstein. Urbana, IL: University of Illinois Press, 1989. 41-53.

⁷Andrew Piper, *Enumerations*, University of Chicago Press, 2019. page 18.

⁸Link to Word2Vec source code: <https://radimrehurek.com/gensim/models/word2vec.html#gensim.models.word2vec.Word2Vec>

⁹The documentation describes `similar_words()` "computes cosine similarity between a simple mean of the projection weight vectors of the given keys and the vectors for each key in the model... correspond[ing] to the word-analogy and distance scripts in the original word2vec implementation." Link to `most_similar()` source

```
woman_similar = model.wv.most_similar(positive = "woman", negative = "man")
```

```
man_similar = model.wv.most_similar(positive = "man", negative = "woman")
```

To make sure that the outputs of this calculation consist of that are most characteristic of each gender, I defined the female term in direct opposition to the male terms. In the code expression, I added parameters specifying the most "positive" and "negative" associations between the genders. The results, then, show words that are most distinctive for each gender, most strongly correlating to "woman" while also being distant from "man" in vector space.¹⁰ I saved these similar words in separate lists called `woman_similar` and `man_similar`, as seen below.

```
woman_similar:
```

```
[('soft', 0.3692586421966553),  
(('named', 0.34212377667427063),  
(('sciatica', 0.3223450779914856),  
(('frilled', 0.3187992572784424),  
(('despaired', 0.31375786662101746),  
(('friend', 0.31238242983818054),  
(('delicious', 0.30853813886642456),  
(('winked', 0.30514153838157654),  
(('notion', 0.3047487139701843),  
(('seductiveness', 0.30290719866752625)]
```

```
man_similar:
```

```
[('chequered', 0.4025157392024994),  
(('fact', 0.3394489586353302),  
(('denounced', 0.3346075117588043),  
(('house', 0.33423593640327454),  
(('curiosity', 0.33144116401672363),  
(('defend', 0.3284823000431061),  
(('dancing', 0.3282632827758789),  
(('marbling', 0.3184848427772522),
```

code: https://radimrehurek.com/gensim/models/keyedvectors.html#gensim.models.keyedvectors.KeyedVectors.most_similar

¹⁰Link to file from this project's source code: <https://github.com/gofilipa/qdr/blob/master/lib/actions.py>

(‘cynosure’, 0.3057470917701721),
(‘rather’, 0.3024100363254547)]

Readers may note that like Underwood’s project, this analysis begins with an oppositional formulation of gender, one that places male and female into a positive/negative axis. However, the goal of this analysis is not to reproduce or reify the binary: rather, it is to bring forth what has been repudiated by each term. Like Butler’s reading of "queer," this analysis finds that the female and male genders are constituted by certain *exclusions* which define the other. In what follows, I read into a selection of these terms, examining them in context, to seek out what constitutes their meaning and, eventually, what must be repudiated in order for them to signify.

5

For example, term "delicious." This term only occurs five times in the novel, and only after Orlando transitions into a woman. Three of these occurrences appear in a single passage, when Orlando is sailing from Turkey back to her native England. In this passage, the ship captain offers Orlando a slice of beef. After politely refusing, Orlando launches into a rapturous speculation on the joys of womanhood.

‘A little of the fat, Ma’m?’ he asked. ‘Let me cut you just the tiniest little slice the size of your fingernail.’ At those words a *delicious* tremor ran through her frame. Birds sang; the torrents rushed. It recalled the feeling of indescribable pleasure with which she had first seen Sasha, hundreds of years ago. Then she had pursued, now she fled. Which is the greater ecstasy? The man’s or the woman’s? And are they not perhaps the same? No, she thought, this is the most *delicious* (thanking the Captain but refusing), to refuse, and see him frown. Well, she would, if he wished it, have the very thinnest, smallest shiver in the world. This was the most *delicious* of all, to yield and see him smile. ‘For nothing,’ she thought, regaining her couch on deck, and continuing the argument, ‘is more heavenly than to resist and to yield; to yield and to resist. (emphasis mine, 114)

6

Delicious, which is repeated insistently, conspicuously, in this passage, describes a distinctly feminine kind of pleasure. In this context, its novelty comes from the opposition to pursuit, here posed as the active form of pleasure proper to the man, that which Orlando enjoyed as a man when he

seduced the Russian princess, Sasha. In contrast to the active pleasure of the pursuer, this passive, feminine pleasure is about withholding and, eventually, submitting. Yet, it is not totally passive: it betrays a kind of power, the power in vascillating between refusal and acquiescence, "to resist and to yield" (114).

7

I run another similarity search, iterating on the term "woman," but this time, with "delicious" as the target word. The top result, the word most related to "delicious" in the text, is "culpable." As it happens, this term occurs in the same scene as above, when Orlando is ruminating on the differences between the sexes on the ship. In this section, Orlando considers what her change in sex means for her sexuality:

And as all Orlando's loves had been women, now, through the *culpable* laggardry of the human frame to adapt itself to convention, though she herself was a woman, it was still a woman she loved; and if the consciousness of being of the same sex had any effect at all, it was to quicken and deepen those feelings which she had had as a man. (emphasis mine, 119)

8

"Culpable," from the Latin "culpa," means fault or blame. In this context, the term modifies the phrase, "the laggardry of the human frame," that is, the human body, "to adapt itself to convention," to the convention heterosexuality, of the female's desire for the male. Like "delicious" above, "culpable" also describes kind of refusal, a refusal to conform to social expectations. But instead of associating refusal with power, it associates it with sexuality. Indeed, this sentence is one of the few in the novel (in all of Woolf's fiction) where homosexuality is considered in explicit terms—"it was still a woman she loved."

The iteration from "delicious" to "culpable" thus models femininity through descriptions of power and passion. If femininity is constituted by these terms, what then constitutes masculinity?

10

Masculinity, as it turns out, is constituted by division. Moving to the terms associated with "man," I begin with the term "chequered," which appears only once, in the first few pages of the story. Here, after describing Orlando's aristocratic lineage ("His fathers had been noble since they had been at all. They came out of the northern mists wearing coronets on their heads"), the narrator paints Orlando stepping into "the yellow pools which

chequered the floor, made by the sun falling through the vast coat of arms in the window" (12). His depiction then veers into lush and exaggerated language.

Orlando stood now in the midst of the yellow body of an heraldic leopard. When he put his hand on the window-sill to push the window open, it was instantly coloured red, blue, and yellow like a butterfly's wing. Thus, those who like symbols [...] might observe that though the shapely legs, the handsome body, and the well-set shoulders were all of them decorated with various tints of heraldic light, Orlando's face, as he threw the window open, was lit solely by the sun itself. 12

11

The image of Orlando's body cut or *chequered* into different colored pieces is indeed a symbol, but one of division rather than nobility. This is a condition exacerbated by the narration, which soon gallops into full-fledged figuration:

Directly we glance at Orlando standing by the window, we must admit that he had eyes like drenched violets, so large that the water seemed to have brimmed in them and widened them; and a brow like the swelling of a marble dome pressed between the two blank medallions which were his temples. Directly we glance at eyes and forehead, thus do we rhapsodize. Directly we glance at eyes and forehead, we have to admit a thousand disagreeables which it is the aim of every good biographer to ignore. 12-13

It bears noting that this scene is the first in many scenes in the novel where the narrator, a self-described "scribe," committed to straightforward description, calls into question his credibility as a biographer. When trying to describe Orlando's beauty, he endows it with mystical qualities—his eyes like flowers, his forehead like marble. The narrator's evocative language undermines his own pretense to objectivity.

12

These momentary slips into figurative language grow into a crisis of signification that recurs persistently through the novel, and entwine with Orlando's own thoughts. Take another term from the list of "woman" vectors, "despaired." Like "chequered," this term occurs in a scene where the line between fiction and reality become blurred. In this scene, Orlando is in the

midst of a period of depression following his abandonment by the Russian princess, Sasha. He is lying beneath his favorite Oak tree, ruminating about the meaning of truth, love, and above all, language itself.

‘The sky is blue,’ he said, ‘the grass is green.’ Looking up, he saw that, on the contrary, the sky is like the veils which a thousand Madonnas have let fall from their hair; and the grass fleets and darkens like a flight of girls fleeing the embraces of hairy satyrs from enchanted woods. ‘Upon my word,’ he said (for he had fallen into the bad habit of speaking aloud), ‘I don’t see that one’s more true than another. Both are utterly false.’ And he despaired of being able to solve the problem of what poetry is and what truth is and fell into a deep dejection. 75

13

Like the narrator, Orlando struggles with the referential qualities of language. First, he attempts plain language, "the sky is blue," "the grass is green." But these prove insufficient to describe a sky of "veils [from] a thousand Madonnas" and grass that "fleets and darkens" like "girls fleeing." Orlando, like the narrator, cannot make language do the work of expression. So he must turn to figurative elements, comparisons of feminine subtlety and evasion which betray his pain from being recently abandoned by a woman. If in the previous scene, Orlando was divided by different colors of light, here he appears divided by doubt about language and its pretention to describe reality.

13.1

He is divided, that is, until he becomes a woman. In chapter six, a scene which mirrors this one under the Oak tree appears to resolve the crisis of signification. Orlando, now a woman, watches a toy boat on the Serpentine river in Hyde Park, London. The toy boat incites a waking vision of Shel, her gender-bending husband, who is a sailor: "as she watched the toy boat climb the ripple, she thought she saw [his] ship climb up and up a glassy wall" (PAGE NUMBER). Her thinking climaxes when she sees the little boat make it to the other side of the wavelet.

‘Ecstasy!’ she cried. ‘Ecstasy! Where’s the post office?’ she wondered. ‘For I must wire at once to Shel and tell him. . .’ And repeating ‘A toy boat on the Serpentine’, and ‘Ecstasy’, alternately, for the thoughts were interchangeable and meant exactly the same thing, she hurried towards Park Lane.

'A toy boat, a toy boat, a toy boat,' she repeated, thus enforcing upon herself the fact that it is not articles by Nick Greene on John Donne nor eight-hour bills nor covenants nor factory acts that matter; it's something useless, sudden, violent; something that costs a life. . . that's what it is—a toy boat on the Serpentine, ecstasy—it's ecstasy that matters. PAGE NUMBER

She hurries to telegram the phrases, "a toy boat on the serpentine" and "ecstasy," to Shel, whom she knows will immediately understand what it means. She realizes that language, its power, is violently ecstatic. If before, under the Oak tree, Orlando struggled to describe reality in plain language, now she has the signficatory power to equate two apparently unrelated terms.

14

An interesting, final example shows how close language and gender are in the novel. Words from both the "woman" and "man" lists of vectors occur within a single passage, the scene where Orlando first meets his beloved Sasha. The words "curiosity," from the "man" list, and "seductiveness," from the "woman" list, appear together in Orlando's tumultuous and agonizing attempt to determine Sasha's gender. Upon seeing Sasha for the first time, cannot tell whether she is a man or a woman:

He beheld, coming from the pavilion of the Muscovite Embassy, a figure, which, whether boy's or woman's, for the loose tunic and trousers of the Russian fashion served to disguise the sex, filled him with the highest curiosity. The person, whatever the name or sex, was about middle height, very slenderly fashioned, and dressed entirely in oyster-coloured velvet, trimmed with some unfamiliar greenish-coloured fur. But these details were obscured by the extraordinary seductiveness which issued from the whole person. Images, metaphors of the most extreme and extravagant twined and twisted in his mind. He called her a melon, a pineapple, an olive tree, an emerald, and a fox in the snow all in the space of three seconds; he did not know whether he had heard her, tasted her, seen her, or all three together. (For though we must pause not a moment in the narrative we may here hastily note that all his images at this time were simple in the extreme to match his senses and were mostly taken from things he had liked the taste of as a boy. But if his senses were simple they were at the same time extremely strong. To pause therefore and seek the

reasons of things is out of the question)... A melon, an emerald, a fox in the snow—so he raved, so he stared. When the boy, for alas, a boy it must be—no woman could skate with such speed and vigour—swept almost on tiptoe past him, Orlando was ready to tear his hair with vexation that the person was of his own sex, and thus all embraces were out of the question.
27-28

Orlando uses seemingly arbitrary metaphors, "A melon, an emerald, a fox in the snow," indicating that at the same time which he cannot place Sasha's gender, he also cannot find the right words to describe her. This dynamic between gender and language is further developed in the narrative form of the passage: as Sasha's gender oscillates between male and female, so the perspective alternates between Orlando's interiority and the narrator's commentary. In particular, the insertion of the narrator's "pause," a pause which purports to explain Orlando's arbitrary metaphors, signals a breach in the diegetic narrative, which reinforces the constructed quality of the scene. By the end, it is fair to say that both gender and language are illustrated as pliant material, one that can be put toward various uses and expressions.

15

Taking words from both the female and male lists of vectors, this passage suggests that the two genders intersect via language's unstable, expressive qualities. Gender, like language, is *fluid*. As a thematic, this fluidity has been well documented by critics: Caughie claims that "Woolf brings out the arbitrariness of [sexual] identity, the arbitrariness of language itself, through Orlando's switching from one sex to the other... as well as through the shifting of her own rhetoric in this novel" (42); Victoria L. Smith asserts that "what happens in the novel... and what it thematizes—language's inability to adequately represent the 'thing itself'—mirrors the undecidability of the text—is it a biography, an autobiography, fantasy, etc." (58). The above passage, with its "switching" and "shifting" discourse, which asserts that word choices are arbitrary and flowing, implies that gender is also a fluid phenomenon.

In this analysis, fluidity is brought forth through close readings of specific terms, "delicious," "culpable," "chequered," "despaired," and in the longer passage, "seductiveness" and "curiosity." For the terms similar to "woman," gender is constituted by a reclamation of feminine refusal into sign of power more than passivity. The masculine gender is constituted through an association with division, which spreads from Orlando's interior experience to the extradiegetic layer of the narration. Orlando's division points to the novel's

larger crisis of signification, which appear to resolve when Orlando becomes a woman. What is divided in the male appears to be resolved in the female. In the end, the repudiated meaning of gender, the meaning which is being brought back through is fluidity.

16

The iterative reading practice brings these meanings of gender to the surface. The lists of word vectors offer starting points for isolating passages throughout the text. As the reader moves from distant to close, from a word vector to its context, the meanings of certain words gain new significations.

1.6 Conclusion

Like *Orlando's* narration, there is something conspicuous about my own presentation of facts. Attentive readers of *Orlando* will know that I left out the second half of the first sentence. The rest of the first sentence, "—was in the act of slicing at the head of a Moor which swung from the rafters" (11)."

Race, and the body, is elided in this analysis.

According to Jamie Hovey, gender and nationality are constructed via race and sexuality. "Sexuality and other dimensions of identity, such as nation and race, are seen as performed through gender" (Hovey 397).

"Beginning with an image of a decapitated Moor, Orlando appropriates the exoticized subjects of colonial discourse to create for its protagonist a sexuality that is not bourgeois or heterosexual. However, Orlando's playful but insistent production of white femininity allows the protagonist to pass as respectable and heterosexual by displacing her transgressive sexuality onto racial others, thereby masking Orlando's masculine-identified literary and sexual desires." (Hovey 398).

2

This elision is because there is a limitation to Butler's theory. Common critiques of Butler point out the limits of this theory for posing gender and sexuality as discursive.

3

According Jay Prosser, Woolf's experimentation with language and narrative form belies the embodied reality of transsexuality. He argues that "Orlando is not about the sexed body at all but the cultural vicissitudes of gender. As h/er narrative propels h/er through four centuries of history, Orlando is free to move beyond h/er body—quite queerly, to break through the limits of the flesh" (Prosser 168). By "the sexed body," Prosser means the physical body that exists in a physical world. Orlando's transgressiveness,

Prosser argues, results from a play of language and literary form that elides the specificity and the lived reality of the “sexed body.”

4

A decade later, Omise'eke Natasha Tinsley writes about the problem of gender fluidity as a metaphor. In her essay, “Black Atlantic, Queer Atlantic: Queer Imaginings of the Middle Passage,” Tinsley argues for the imbrication of sexuality and race through the lens of the Black Atlantic.

Her critique re-works the trope of fluidity which, drawing from the ocean, “is not an easy metaphor or queer and racially hybrid identities but for concrete, painful, and liberatory experience” (192-193). For Tinsley, fluidity is an opportunity for “a return to the materiality of water to make its metaphors mean more complexly, shaking off settling into frozen figures” (212). Tinsley theorizes fluidity as a “social liquidation,” being stripped by the water, particulars of identity washed away in the current. She explains that “brown bodies are gender fluid not because they choose parodic proliferations but because they have been ‘washed of all this lading, bag and baggage’” (209).

This is taken up in chapter four on the flesh.